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Chapter 1

What is PerSense?

What does PerSense do?

Introduction and Overview

PerSense is a tool for economic analysis. Its six data sheets are so flexible and general in their format that almost any financial question you can imagine fits logically and neatly into one of them.

There are three modes for working with *PerSense*. In the simplest, you work from a script: the program talks you through your problem, requesting input as you go. For more complex problems, choose from more than forty examples on screen, and substitute your numbers for the example numbers.

In its most powerful mode, *PerSense* is a completely flexible calculator. Your job is to place the numbers you know in appropriate rows and columns of one of *PerSense*'s data sheets. *PerSense* knows when you have provided sufficient information to solve the problem, and fills in the answers as you press **Enter**.

Your input numbers are "hard" data, displayed light-on-dark on the screen; *PerSense*'s output is "soft" data, displayed dark-on-light. Soft data is erased and recomputed each time you make a change.

Behind the scenes, *PerSense*'s intelligence is at work, analyzing the combination of data you have entered, determining what quantities can be computed, selecting the appropriate algorithms or formulas, executing them, and routing the answers to the appropriate cells on your screen. You have only to read the output.

Every economic decision that we make involves the trade-off of economic costs and benefits. Sometimes there are intangible elements involved – uncertain outcomes and unknown probabilities. But often a great part of the analysis involves analyzing cash streams at different times. One of the things *PerSense* does best is to reduce complex cash flows at different times to common terms so that they can be compared directly.

1. Introduction

Investment analysis

Some applications in this category include:

- Analyzing annuities and other payment structures.
- Valuing legal claims for damages.
- Financial planning
- Investment analysis and internal rate of return (IRR)
- Comparison of business options – which is the best deal?
- Deciding whether to lease or to purchase

In *PerSense Pro*, all of the power of this investment analysis system can be merged seamlessly with actuarial tables. Payments that continue for life, and amounts payable on death can then be included. This makes *PerSense* ideal for comparison of life insurance and annuity investments with other, more conventional investment instruments.

Actuarial analysis

Loans

A third major focus of *PerSense* involves loans. Many programs will calculate a payment amount for regular monthly installment loans. *PerSense* will, in addition, allow you to structure a loan to meet the parties' particular needs:

- Multiple balloon payments
- Interest-only for a period of time
- Fixed payment to principal
- Skipped payments
- Combinations of monthly, weekly, and quarterly payments for the same loan

Loan interest has its own world of rules, and *PerSense* can be configured to emulate all of the major and minor standards that are commonly applied:

- 360- or 365-day basis (The 365/360 variant is also supported.)
- Points applied at settlement
- Monthly, weekly, or daily rates
(also quarterly, biweekly, bimonthly, semi-monthly, semi-annually)
- Canadian rates
- Variable rate loans
- American rule (simple interest)
- Rule of 78
- Interest in advance or arrears
- Prepaid interest at settlement

PerSense will calculate loan payments with any combination of these options, and then print an amortization schedule, with each payment divided into interest and principal parts. Federal law requires that lenders report true interest rates to the borrower in the form of an APR (annual percentage rate); *PerSense* will calculate an APR for the most complex scenario you might imagine, with points, irregular payments, multiple rate changes and balloon payments.

Amortization tables

Professional product at a mass-market price

Mortgage screen

Amortization screen

Investment screen

Present Value screen

1. Introduction

PerSense is a professional tool, on the cutting edge of computational capabilities in several distinct areas. Yet its face is so simple and logical that thousands of attorneys, executives, and consumers have found it accessible and valuable. Because it has been designed and marketed for high sales volume, it is being sold for a fraction of the price of many "professional" products that contain only a subset of *PerSense*'s complete array of computational capabilities.

PerSense provides six different computation screens, each of which specializes in one type of computation. The most common types of interest computations can be done on any of the six screens, but for special applications you will find that choosing the appropriate screen helps you to visualize and structure your problem.

The Mortgage screen provides the simplest access to common, straightforward loan computations. In addition, tools are available to explore and compare several different loan scenarios.

The Amortization screen is an extremely flexible and powerful loan calculator. Its Standard mode handles any loan amortized with regular payments and a single interest rate, computes APR with points charged, tells you instantly your balance on any date in the future, and generates amortization tables. Extended mode (press **ⓧ**) permits changes in interest rate, balloon payments, fixed payments to principal, interest-only payments, and negative amortization.

The Investment screen is for financial planning and investment evaluation. As a planning tool, it balances deposits against withdrawals, and solves for any one dollar amount. If you like, you may use inflation-adjusted *constant* dollars throughout. Actuarial probabilities can be incorporated in this analysis for retirement planning or evaluation of insurance benefits (*Pro* version only).

The Investment screen will also evaluate investment performance. For example, you may have added to your mutual fund account periodically, and sold shares occasionally. Simply by listing all **Funds In** and **Funds Out**, you can ask *PerSense* to calculate an overall rate of return.

The structure of the Investment screen is similar to that of the **Present Value** screen, and their functions overlap considerably. In fact, the question whether to work in the Investment screen or the **Present Value** screen is usually a matter only of convenience and intuitive clarity; both will yield identical results.

The structure of the **Present Value** screen is based on the concept of valuation: reducing any payment stream to its value at any one time. This kind of analysis is common in civil law and financial analysis.

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However, the paradigm is a very powerful one, and this screen can be used in many other ways. Computation of an internal rate of return (IRR) can be performed as the "inverse" of a present value computation on this screen. Financial planning and investment analysis can be performed here, although these functions are most at home on the **Investment** screen.

Press **Ctrl-A** for an actuarial overlay, in which probabilities of survival can be seamlessly integrated into any of the above computations, extending the notions of present value and IRR (*Pro* version only). With this extension, the **Present Value** screen can evaluate annuities, pensions, and death benefits, and compare any of these with traditional investment instruments.

The **Arbitrary Loan** screen allows you to structure your own loan from the ground up, with no presumptions about regular payments or even regular interest rates. Rates may be changed as often as you like. Payments may occur at any time, in any amount. New lending may be recorded simply as a negative payment.

Sometimes a loan agreement provides for regular monthly payments, but the lender in fact wants to record the exact day and amount of each payment, and charge exact daily interest on the principal. The **Arbitrary Loan** screen is the ideal tool for this task, and indeed it offers six different options for ways to calculate interest in this non-standard situation.

The model here is a single line for each "event": a deposit or a series of deposits or a withdrawal or rate change. This is the natural place for accounts payable, passbook savings, and credit card accounts. You can also calculate the time when periodic deposits will accumulate to a target sum, or the date on which a loan will be completely paid off.

Many common computations can be done on more than one screen. Straightforward computations such as monthly payment, given loan amount, term, and rate, can be performed on five of the six screens. As you become more familiar with *PerSense*, you will develop your own favorite methods for structuring the situations you typically handle.

Two types of **Help** windows are available within the program.

The function key **F1** brings you *contextual help*, that is, the program tries to guess, based on context, what topic you are looking to explore. If the **Help** text is not exactly what you need, you can use hypertext features to reach other help topics: click the mouse on any highlighted word which points in the right direction or, if you are not using a mouse, tab the highlight bar to that word and press **Enter**. Select the word **INDEX** (in the upper right of every **Help** screen) to choose a word or phrase from the help index. Press **Esc** to leave the help system and return to your work where you left it.

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Most people learn best by example, and for this purpose, a second **Help** system is available. Press **Alt-F1** from any of the six calculation screens to bring up a list of sample computations for that screen. Select an example that is similar to the computation you wish to perform. You will be offered a plain English description of the problem, together with instructions for setting it up. At the same time, enough of the computational screen will remain active for you to do the example right there and then. You can change the numbers to fit your own situation, and recalculate, all without leaving **Example** mode.

Many users will have little need for this manual, because the major *PerSense* commands are so intuitive and natural. Basic data entry is similar to working within a spreadsheet. Use the mouse or the cursor keys, together with **Tab** and **Shift-Tab**, to move about the screen. Computation is performed and the display is updated each time you press the **Enter** key, or press letter **C** (or **F9**) to force recalculation. Explore the pull-down menus along the top of the screen by pressing **Esc** (or **F10**) and selecting an item with the cursor keys.

This manual contains most of the examples available in **Example** screens, as well as many additional ones that are more complex or obscure.

Chapter 2 provides the instructions you will need to install and use the software, manage files, and transfer data to other programs.

Chapter 3 is a Quick Tour of the program, with a description and example from each of the six screens.

Chapter 4 discusses *PerSense*'s **Settings** menu, where you can specify computational standards, and adapt *PerSense* to your computer and printer.

Chapters 5 through 10 document each screen's options in detail and provide numerous examples. If you have any doubt about how to structure your interest-related question in *PerSense*, you will find a similar calculation here to use as a model.

Chapter 11 offers instructions for three *PerSense* accessories: a built-in calculator, a calendar, and a mini-word processor.

A Glossary at the back of the manual explains some technical terms commonly used in banking and finance.

An Appendix briefly describes the computational methods on which *PerSense* is built.

About this manual

Arbitrary Loan screen

Chronological screen

Help